Preparing for the Exceptional: Extreme Climate Profiles for Planning and Vulnerability Assessment

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When planning around typical conditions isn't enough

The energy sector currently uses Typical Meteorological Year (TMY) climate profiles for building design, peak demand, production estimates, and asset planning. The TMY climate profile is one year of hourly data that represents historical or future median meteorological conditions for a point location.

However, ***climate profiles built around typical conditions may be insufficient at representing the types of future events that will drive peak electricity demand***. Cal-Adapt has developed new Extreme Meteorological Year (XMY) profiles to represent the types of future events that will drive peak electricity demand.

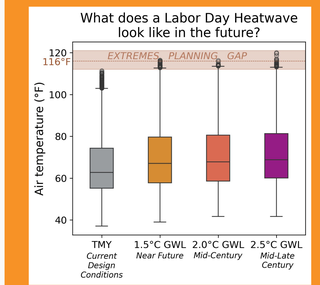


Figure 1. California's Labor Day 2022 heatwave produced record-breaking electricity demand exceeding 52,000 MW.

Boxplot distribution of temperatures for Sacramento comparing typical standards to extreme heat tailored future conditions at several planning horizons.

Cal-Adapt collaboratively developed the **first of its kind, open-source, scientifically robust extreme hourly profiles** for grid stress tests and planning.

Developed for & with users

Extreme Climate Profiles on the Cal-Adapt were co-produced with Electric Power Research Institute (EPRI) and Argonne National Laboratory (ANL). Together, we have identified scientific best practices, relevant data, methods, and workflows that both inform and utilize climate profiles, directly supporting energy-sector planning needs.

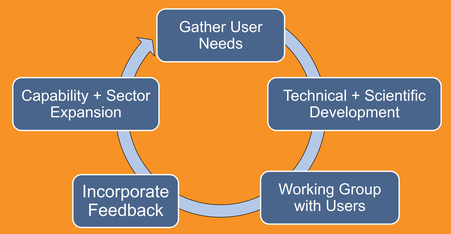


Figure 2. The production cycle to develop hourly climate profiles for the *Cal-Adapt: Analytics Engine*. This work is iterative and on-going.

Cal-Adapt offers extreme profile options to meet today's planning needs

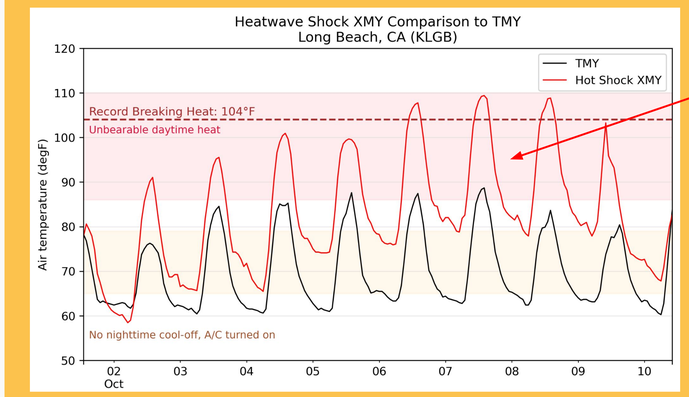


Figure 3. Comparison of typical conditions to extreme heat conditions for Long Beach, CA during heatwave conditions at *present-day* (1.2°C).

Recent extreme heat waves have **surpassed historic temperature records**:

- **Oct 2026 Heatwave**: Downtown LA reached 104°F, breaking a 1953 record → *running 15-22 degrees F above typical conditions*
- **Labor Day 2022 Heatwave**: Temperatures reached 116°F in Sacramento → *CAISO required to institute blackouts due to record-breaking electricity demand, estimated to be a 1-in-450 year event*
- **Aug 2020 Heatwave**: Temperatures reached 121°F in Los Angeles County (highest ever) → *rolling outages*

A ***typical*** TMY profile would not be able to represent extreme events such as these heatwaves because they are designed to reproduce median conditions. However, an ***extreme*** XMY profile can be used to plan for an event like these heatwaves.

Cal-Adapt Extreme Profiles better **support extreme heat planning** when compared to typical conditions

Ready-to-go solutions

Cal-Adapt carefully considered ***what kind*** of extreme characterization may be required for your planning and solutions. Extreme weather is not a single hazard and no single extreme climate characterization can adequately represent the **diverse conditions** that drive community and system risk. Because planning objectives range from peak demand events to prolonged heatwaves, coldsnaps, and compound events, relying on a "*one-size-fits-all*" approach can obscure critical vulnerabilities and lead to incomplete assessment.

I need to plan for rare, high impact extreme heat

*Use a **Shock XMY**!*

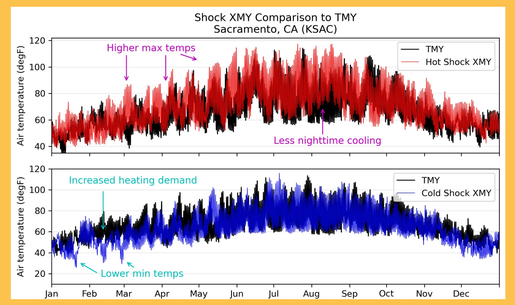


Figure 4. Hot Shock XMY (Top) or Cold Shock XMY (bottom) profile compared to the TMY for the Sacramento Executive Airport at mid-century (2.0°C GWL). The Shock XMYs have more extreme hot or cold snaps relative to the TMY.

I need to perform a specific stress test

*Use a **Persistence XMY**!*

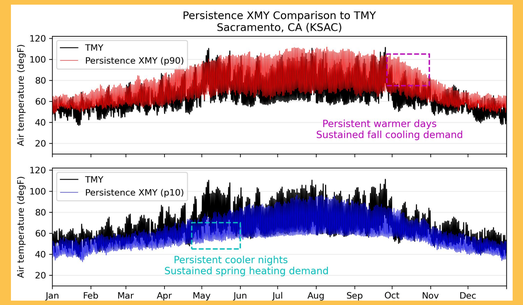


Figure 5. Hot (Top) and Cold (bottom) Persistence XMY profile compared to the TMY profile for Sacramento Executive Airport for the 90th or 10th percentile extreme at mid-century (2.0°C GWL). The Persistence XMYs have more sustained periods of heating or cooling demand compared to the TMY.

Where do you need to plan for extremes?

Vulnerability Assessment

Vote here!

Resource Planning

Vote here!

Emergency Preparedness

Vote here!

Long-Term Adaptation

Vote here!

Operations & Decision-Making

Vote here!

Policy & Investment

Vote here!

Download the Climate Profiles now!

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